



Information technology, operational, and management competencies for supply chain agility: Findings from case studies

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ARTICLE INFO

Article history:

Available online 24 December 2010

Keywords:

Supply chain agility
IT competence
Operational competence
Management competence
Supply chain competence
Case studies

ABSTRACT

Given the growing emphasis on the need for supply chain agility to sustain competitive advantage, this study explores the impact of the relationship between supply chain competence and supply chain agility on firm performance. Although the influence of supply chain agility on firm performance has been acknowledged, this study further articulates the relationship from the perspective of inter-organizational collaboration. We develop a conceptual model based on the resource-based view and employ a multi-case study method in this exploratory research. Our findings highlight the importance of distinguishing the difference between supply chain agility and supply chain competence and their impact on firm performance. This study contributes to the growing body of conceptual and empirical literature on supply chain agility and adds to the understanding of the complexity of supply chain competence.

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1. Introduction

Currently, business environment is characterized by pressure caused by stiff competition, constant changes (e.g., product/technological innovations, decreasing product lifecycles, and product proliferation), and a high level of market uncertainty and unpredictability (Braunscheidel and Suresh, 2009; Overby et al., 2006; Swafford et al., 2006a,b; White et al., 2005). Choi (2007) and Christopher (2000) show that demand uncertainty and limited visibility have long been a challenge to supply chain management. Such challenges require organizations to respond promptly to market changes to sustain competitive advantage and business success (Sambamurthy et al., 2003; Swafford et al., 2006a). Supply chain agility is the organization's ability to respond to unexpected market changes and convert these changes to business opportunities (Swafford et al., 2008). Thus, it is one of the most essential elements helping an organization survive a turbulent and volatile environment (Agarwal et al., 2007; Bottani, 2009; Braunscheidel and Suresh, 2009; Lin et al., 2006; Oosterhout et al., 2006; Overby et al., 2006; Sarker et al., 2009; Swafford et al., 2006a, 2008).

An emerging body of management research focuses on understanding the importance of supply chain agility in coping with the challenges in a business environment. Within the domain of research, most of the articles focus on identifying the enablers or antecedents of supply chain agility. The potential enablers or antecedents include collaborative relationships with supply chain partners (Braunscheidel and Suresh, 2009; Khan and Pillania, 2008; Lin et al., 2006; White et al., 2005), supply chain flexibility (Braunscheidel and Suresh, 2009; Swafford et al., 2006a, 2008), internal integration (i.e., information and/or process) (Braunscheidel and Suresh, 2009; Lin et al., 2006), and information technology (IT) (Braunscheidel and Suresh, 2009; Swafford et al., 2008; Sambamurthy et al., 2003; Li et al., 2006; Fink and Neumann, 2007). Despite various studies, there is no research providing a comprehensive framework for supply chain competence for enhancing supply chain

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agility. In contrast, existing research on supply chain agility mainly focuses on internal supply chain and its immediate entities (Braunscheidel and Suresh, 2009; Swafford et al., 2008; White et al., 2005). This gives rise to the research concern on the role of inter-organizational collaboration in the relationship between supply chain competence and agility.

With regard to the emerging business value of supply chain agility and insufficiency of research, this study aims to develop a framework for supply chain competencies, particularly for supply chain agility and firm performance. Specifically, this study aims to uncover the answers to the following research questions:

- a. What are the entities that encompass supply chain competence supporting supply chain agility?
- b. How do information technology, operational, and management competencies enable supply chain agility?
- c. How does supply chain agility enhance firm performance?

With reference to resource-based view (RBV), this study develops a conceptual model and derives strategic implications for the use of competencies as strategy for managing supply chain agility. This study hopes to provide organizations with guidelines for understanding the competencies crucial to sustaining competitive advantages and surviving in a hyper-competitive environment. This paper proceeds as follows. In the theoretical background, the concepts of supply chain agility and supply chain competence are discussed. Supply chain competence is conceptualized as having three conceptual components: IT competence (IT integration and flexibility), operational competence (supply chain integration, flexibility and learning orientation), and management competence (top management role and vision and employee competence). A conceptual framework that explains the causal relationship among supply chain competence, supply chain agility, and firm performance is then presented. Subsequently, three case studies on the textile and fashion industry are presented to validate the proposed research model using a case study approach. This research builds a theoretical foundation for instrument development and tests the individual causal relationship among the above-mentioned constructs.

2. Supply chain agility

The concept of agility was introduced by researchers of the Iacocca Institute of Lehigh University in 1991 (Bottani, 2009). It was first introduced within the field of manufacturing and in the concept of flexible manufacturing systems (Sarker et al., 2009; Sarker and Sarker, 2009; White et al., 2005). Since then, it has received increasing attention from both academic and commercial fields. In the field of management information system, supply chain agility is a relatively new concept (Braunscheidel and Suresh, 2009). Researchers conceptualize supply chain agility as a broad and multi-dimensional construct. In the most general sense, supply chain agility is defined as the capability of supply chain functions to provide a strategic advantage by converting unexpected market uncertainties and potential and actual disruptions into competitive opportunities through assembling requisite assets, knowledge, and relationships with speed and surprise (Bottani, 2009; Braunscheidel and Suresh, 2009; Christopher, 2000; Khan and Pillania, 2008; Sambamurthy et al., 2003; Swafford et al., 2006a). On one hand, supply chain agility encompasses the exploration and exploitation of market opportunities. On the other hand, it encompasses the ability to deliver innovative products and services in a timely and cost-effective manner (Overby et al., 2006; Sambamurthy et al., 2003; Sarker et al., 2009; Swafford et al., 2006a; Zhang and Sharifi, 2000). Supply chain agility requires an organization to supervise closely the legally separate but operationally interdependent parties, such as suppliers, manufacturers, and distribution, to maintain a close and coordinating relationship (Lin et al., 2006; Yusuf et al., 2004). It enables an organization to reduce manufacturing costs, enhance customer satisfaction, remove non-value added activities, and therefore maintain a competitive position in a competitive environment (Bottani, 2009; Braunscheidel and Suresh, 2009; Lin et al., 2006; Narasimhan et al., 2006; Swafford et al., 2006a).

RBV has been used to understand the conditions for sustaining competitive advantages (Ray et al., 2005). It posits that the organizational resources are the predictors of organizational performance (Kearns and Lederer, 2003). RBV provides a robust framework for analyzing the relationship between supply chain competence and firm performance. To compete amid the uncertainty and changes in a business environment, an organization should develop core competencies and capabilities that may help it survive the competitive environment. Zhang et al. (2003, p. 176) posits that “competence emphasizes technological and production expertise at specific points along the value chain while capabilities are broadly based and encompass the entire value chain.” They argue that capabilities are visible to the customers, whereas competencies are internal and support these capabilities. Following the above argument, this study proposes that supply chain agility is the capability of an organization to respond to market changes visible to customers using a set of supply chain competencies that enable such capability.

This study argues that supply chain agility is rare, valuable, and imperfectly imitable, and may therefore result in superior, long-term performance. Supply chain agility is not about rules and procedures, which can be easily implemented or imitated; rather, it is about a complex philosophy of coordination and integration among different parties and functions along the value chain. Supply chain agility helps sustain competitive advantage by continually sensing market changes and quickly responding to these changes. In relation to RBV, this study argues that supply chain agility is an organizational capability that enables an organization to sustain competitive advantages. Therefore, this study proposes the following:

Proposition 1. *Supply chain agility is positively associated with firm performance.*

3. Supply chain competence

Supply chain competence refers to the framework for technological, production, and management expertise that supports supply chain capabilities. To develop supply chain agility, an organization has to possess a number of supply chain competencies. Supply chain agility requires different parties and functions of supply chain to maintain a close and coordinated relationship, allowing an organization to respond to market changes in a timely and cost-effective fashion. Thus, the supply chain operation must be sufficiently competent to commit to a new course of action in a timely and cost-effective manner. IT and human resources are two essential elements that support the operation of the supply chain. Without the support of IT, the flow of information and resources becomes much slower than required to enable a timely response to market changes. Top management and operational personnel should be equipped with the ability to analyze market and operational information and give directions to the supply chain to meet market demands. This study argues that IT, operational, and management competencies are three inter-supportive competencies for enabling supply chain agility. In this study, supply chain competence is conceptualized as a latent construct that encompasses three essential dimensions: IT, operation, and management. Each of these dimensions reflects the ability to perform supply chain agility. Interviews with directors of the three cases also confirm the conceptualization of supply chain competence. The framework for supply chain competence for supply chain agility is presented in Fig. 1.

The definition of the major constructs is summarized in Table 1. In the following sections, the effect of each supply chain competence on supply chain agility is explained. The research framework is presented in Fig. 2.

3.1. Operational competence

Operational competence refers to the ability of an organization to use its resources to facilitate supply chain capability, particularly supply chain agility. This study conceptualizes operational competence as a second-order latent construct composed of sub-constructs, namely, supply chain integration, supply chain flexibility, and supply chain learning orientation. These sub-constructs are chosen based on the criteria of supply chain agility (i.e., the need for proper response to market change, exploiting changes, and taking advantages of changes as opportunities) (Zhang and Sharifi, 2000). They help (1) maintain an efficient flow of information and materials among different parties and functions along the supply chain; (2) reserve a wide variety of operational resources that can be activated quickly; and (3) maintain continuous search of knowledge to improve supply chain efficiency and responsiveness. We argue that the effects of the chosen sub-constructs help an organization fulfill the criteria for supply chain agility. First, efficient flow of information helps organization to keep track of market needs and foresee possible changes in which action is required to sustain competitive advantages. Second, efficient flow of materials allows an organization to relocate resources in order to respond promptly to market changes. Finally, continuous search of knowledge creates an organizational culture that views changes as opportunities for improvement. Thus, the extent to which an organization masters the above-mentioned competence will certainly provide a great impact on supply chain agility.

3.1.1. Supply chain integration

Supply chain integration has been regarded as one of the most important competencies in supply chain management (Pearcy and Giunipero, 2008). It is defined as the formation of a network in which separate supply chain partners collaboratively manage intra- and inter-organizational processes to arrive at mutually acceptable outcomes. (Kim and Cavusgil, 2009; Myler and Broadbent, 2006; Richey et al., 2009; Wong and Boon-Itt, 2008; Zailani and Rajagopal, 2005). It aims to achieve effective and efficient flows of information, products and services, resources, and cash to provide maximum value to the customer at low cost and high speed (Flynn et al., 2010). With an integrated supply chain, a high degree of operational

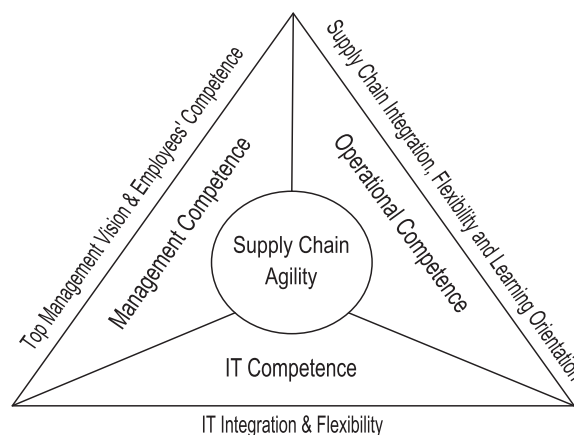


Fig. 1. Framework of supply chain competence for supply chain agility.

Table 1
Definition of major constructs.

Construct	Definition	References
Supply chain competence	Framework for technological, production, and management expertise supporting supply chain capabilities	N.A.
IT competence	Extent to which a firm is knowledgeable about and effectively utilizes IT to manage information within the firm	Tippins and Sohp (2003)
Operational competence	Ability of an organization to use its resources to facilitate supply chain capability, particularly supply chain agility	N.A.
Management competence	Ability of an organization to use its human resources to facilitate supply chain capability, particularly supply chain agility	N.A.
Supply chain agility	Capability of supply chain functions to provide a strategic advantage by converting unexpected market uncertainties, as well as potential and actual disruptions, into competitive opportunities by assembling requisite assets, knowledge, and relationships with speed and surprise	Khan and Pillania (2008); Sambamurthy et al. (2003); Christopher (2000); Bottani (2009); Swafford et al. (2006a); Braunscheidel and Suresh (2009)
IT integration	Extent to which information systems are linked and information is shared among different functions and supply chain parties, thereby effectively creating a virtual supply chain	Lin et al. (2006), Rai et al. (2006)
IT flexibility	Ability of IT infrastructure to adapt to both incremental and revolutionary change in the business or business process with minimal penalty to current time, effort, cost, or performance	Nelson et al. (1997), Conboy (2009)
Supply chain integration	Formation of a network in which separate supply chain partners collaboratively manage intra- and inter-organizational processes to arrive at mutually acceptable outcomes	Kim and Cavusgil (2009), Wong and Boon-Itt (2008), Richey et al. (2009), Zailani and Rajagopal (2005), Li et al. (2006)
Supply chain flexibility	The various states a manufacturing system can adopt; the ability to move from the production of one product to another, and the ability to perform satisfactorily by coming up with good quality of products within a specified range	Stevenson and Spring (2007)
Supply chain learning orientation	Development of new knowledge or insights and using knowledge that has the potential to influence supply chain capability and enhance competitive advantage	Rattanaphaphtham and Ussahawanitchakit (2008)
Top management role and vision	The ability of an organization's management to value and directly appropriate organizational response to market changes	N.A.
Employees' competence	The ability of an organization's participants to implement the organizational response to market changes	N.A.

visibility and inter-organizational cooperation (Rosenzweig et al., 2003) is foreseen. When sudden changes occur in the market, the organization can cooperate strategically with supply chain partners and respond to market changes in a cost-efficient manner. On one hand, prior research has proposed or proven a direct positive relationship between supply chain integration and business performance (Fabbe-Costes and Jahre, 2008; Flynn et al., 2010; Kim, 2009; Knapp et al., 2006; Li et al., 2009; Narasimhan and Kim, 2002; Rosenzweig et al., 2003; Zailani and Rajagopal, 2005). On the other hand, recent research has shown that supply chain integration is indirectly related to business performance through competitive capabilities such as customer services, efficiency, and effectiveness-related capability (Chen et al., 2009; Kim, 2009; Rosenzweig et al., 2003). Although the above-mentioned capabilities are proven to be related to the concept of supply chain agility in certain aspects, no research has explored the linkage between supply chain integration and supply chain agility. Therefore, we posit the following:

Proposition 2a. *Supply chain integration is positively associated with supply chain agility.*

3.1.2. Supply chain flexibility

Flexibility is one of the most prominent antecedents of agility (Conboy, 2009). In the domain of supply chain study, supply chain flexibility pertains to the various states a manufacturing system can adopt, the ability to move from the production of one product to another, and the ability to produce products of good quality within a specified range (Stevenson and Spring, 2007). According to Swafford et al. (2008), supply chain flexibility can be classified into strategic flexibility and manufacturing flexibility. Strategic flexibility is the competence to identify changes in the environment, commit resources quickly to new courses of action in response to change, and recognize and act promptly when it is time to halt or reverse such resource commitments (Shimizu and Hitt, 2004). Manufacturing flexibility is the competence to manage manufacturing resources to meet various customer requests (Zhang et al., 2002, 2003). Based on the definition, strategic flexibility is considered related to prompt decision and commitment to respond, whereas manufacturing flexibility is related to operational ability to implement a strategic decision.

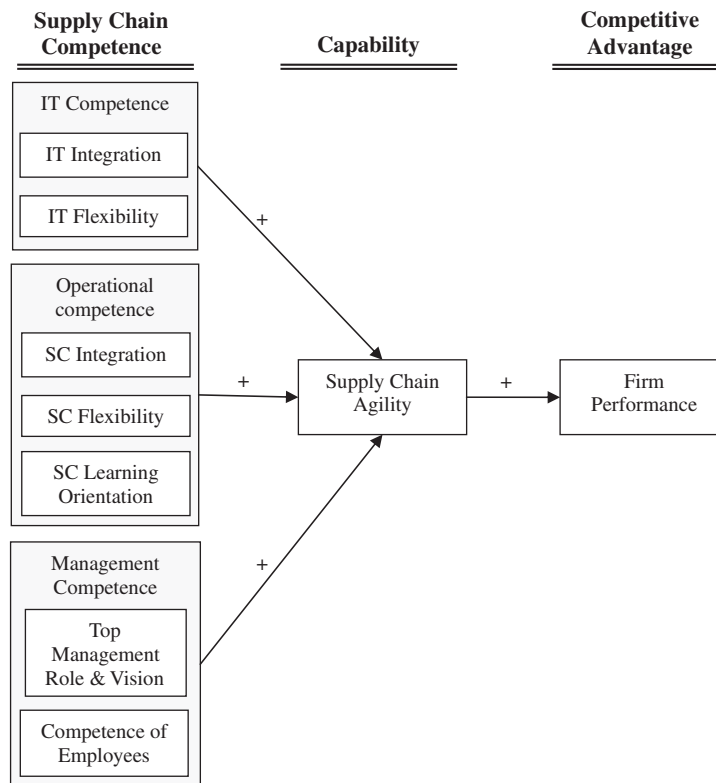


Fig. 2. Research framework.

Through supply chain flexibility, an organization may promptly and strategically reverse its actions and produce different combinations of products in a fast and cost-efficient manner. This is accomplished through the ability of infrastructure and labor resource to perform different operations and the ability of the materials and procedural frameworks to produce different products and services. Swafford et al. (2006a, 2008) find an empirically significant relationship between internal supply chain flexibility and agility. This study further extends the focus to the role of inter-organizational collaboration in enabling supply chain flexibility and tests the proposition again using three case studies. Based on the preceding discussion, we offer the following proposition related supply chain agility:

Proposition 2b. *Supply chain flexibility is positively associated with supply chain agility.*

3.1.3. Supply chain learning orientation

Agility requires the organization to respond promptly to unexpected market changes. Therefore, it is important for an organization to learn continually and prepare well for any challenge that may come up in a turbulent environment. Learning orientation is composed of (1) commitment to learning, which refers to the value placed on learning; (2) open-mindedness, which pertains to the willingness to evaluate critically the supply chain operations and accept new ideas; and (3) shared vision, which relates to the consensus on direction and focus of learning among organizational members (Braunscheidel and Suresh, 2009; Rattanaphaphtham and Ussahawanitchakit, 2008). Organizations with a high degree of learning orientation are able to learn about customers, competitors, and market situation, allowing them to have a better chance at sensing and acting upon unexpected events and trends in the marketplace (Tippins and Sohp, 2003). Moreover, organizational learning prepares an organization with the most updated knowledge on systems, procedures, technologies, and benchmarking to use in satisfying market changes in a quick and cost-efficient manner. Therefore, this study proposes that supply chain learning orientation enhances supply chain agility.

Proposition 2c. *Supply chain learning orientation is positively associated with supply chain agility.*

3.2. Information technology competence

IT has facilitated the formation of global supply chain markets; changed the relationships among customers, manufacturer, and suppliers; and accelerated and visualized the product, financial, and information flows that need to be accurately

and promptly delivered to reflect the status in the supply chain. IT competence provides fundamental support to supply chain agility. It is defined as the extent to which a firm is knowledgeable about and effectively utilizes IT to manage information within the firm (Tippins and Sohp, 2003). IT plays an essential role in enabling the sensing and response capabilities of an organization (Fink and Neumann, 2007; Lin et al., 2006; Overby et al., 2006; Sambamurthy et al., 2003). Examples of IT that enhance supply chain capability are (1) mobile and wireless technologies [e.g., radio frequency identification (RFID) and sensor network technologies] for real-time data collection; (2) integration technologies (e.g., extensible markup language (XML) and Web services) for real-time monitoring of events through portals; and (3) business process re-engineering and management tools for business process automation and redesign of supply chain system or enterprise system. Synchronizing all the essential IT activities to achieve supply chain agility is crucial. Top management should actively engage in a strategic IT plan for supply chain agility and measuring its effect on supply chain performance. The researchers of this study are convinced that the business value of IT competence is manifested in its contributions to supply chain responsiveness. It enables a supply chain with a high degree of visibility, connectivity, responsiveness, and flexibility. Among different IT competencies, IT integration and IT flexibility are deemed to provide the most obvious contributions to supply chain agility.

3.2.1. IT integration

Information is one of the most important resources powering global supply chain competitiveness. IT integration is defined as the extent to which information systems are linked and information is shared among different functions and supply chain parties, thereby effectively creating a virtual supply chain (Knapp et al., 2006; Lin et al., 2006). IT integration helps coordinate supply chain functions and partners through the sharing of information related to demand forecasts, production schedules, inventory, and production quality that dictate supply chain activities (Li et al., 2009). Moreover, IT integration facilitates knowledge sharing along the supply chain. IT integration also includes the exchange of knowledge with partners up and down the supply chain, allowing them to collaborate and to create synchronized replenishment plans.

IT integration provides efficient, timely, and transparent business information to the appropriate parties and reduces the time needed to share knowledge and information (Pearcy and Giunipero, 2008). Therefore, IT integration reduces response time to unforeseen events and market changes (Angeles, 2009; Mondragon et al., 2004; Swafford et al., 2006b). IT integration does not only lead to a smooth flow of information, it also provides significant support to the flow of physical resources and cash. With an integrated information system, different parties along the supply chain are able to access the operational information of other functions or departments. For instance, suppliers may be able to access the inventory information of their customers and prepare for stock delivery on time. Moreover, when a manufacturer needs to meet a sudden increase in demand, a well-integrated IT system can provide a clear picture of the supply chain status, inventory status (of the manufacturers or its suppliers), and even the service capability of its logistic service providers. The manufacturer can immediately grab the complete picture of the supply chain from the suppliers to the distributors and calculate how long it would take to respond to market changes. Therefore, a more efficient resource flow, with the help of IT integration, is foreseen. Efficient flow of information and materials helps organization to keep track of market needs and allows the organization to relocate resources in a responsive manner. Although Swafford et al. (2008) do not find a significant relationship between IT integration and supply chain agility, we reason that their study focused only on internal supply chain. This study extends the focus to the inter-organizational IT integration and investigates the way it enhances supply chain agility. Inter-organizational IT integration facilitates better supply chain agility by enabling real-time information exchange and sharing among supply chain partners. Therefore, we are convinced that there is an association between IT integration and supply chain agility. We believe better IT integration enables supply chain agility capability. Therefore, we posit:

Proposition 3a. *IT integration is positively associated with supply chain agility.*

3.2.2. IT flexibility

IT flexibility is defined as the ability of IT infrastructure to adapt to both incremental and revolutionary changes in the business or business process with minimal penalty to current time, effort, cost, or performance (Conboy, 2009; Nelson et al., 1997). It is characterized by (1) connectivity, which refers to the ability of the information technology component to attach to other components within the organization or with other organizations; (2) compatibility, which pertains to the ability to share information across any information technology component within the organization or with other organizations; and (3) modularity, which denotes the ability to add, modify, and remove information technology components with ease and without negative effect on performance (Byrd and Turner, 2000; Fink and Neumann, 2009).

IT flexibility provides support to the organizational supply chain to prepare for unexpected market changes and conditions without increased cost or time (Byrd and Turner, 2000). For instance, it allows the organization to easily integrate new technological components, such as RFID, with existing IT infrastructure. IT flexibility allows the organization to enhance its IT infrastructure with ease and in a cost-efficient manner based on the market situation, and enables the organization to respond to market changes at the most appropriate time. The ease of modifying the technology component in existing IT system equips the organization with the competence to advance the supply chain operation without affecting performance. This valuable competence can support any need for change in the supply chain system in response to market changes. Therefore, IT flexibility enables the development of better supply chain agility.

Proposition 3b. *IT flexibility is positively associated with supply chain agility.*

3.3. Management competence

Although IT and operational competencies provide a strong foundation for supply chain agility, supply chain agility cannot be fully actualized without management competence. Unlike a traditional business model, supply chain agility requires a high degree of inter-organizational cooperation and collaboration. This study explores the effect of such management competence, that denotes the ability of an organization to use its human resources to facilitate supply chain capability, in achieving a high degree of supply chain agility. There is no article analyzing the effect of management competence on supply chain agility. Therefore, we can only refer to the literature on “organization agility” (Bottani, 2009; Fink and Neumann, 2007) and identify the role and vision of top management and competence of employees as the enablers of supply chain agility.

3.3.1. Role and vision of top management

Role and vision of top management infers the ability of an organization’s management to value and directly appropriate organizational response to market changes. Top management plays an important role in driving value creation through supply chain management. Supply chain agility requires the organization to identify market changes and commit resources quickly to new courses of action or reverse such resource commitments (Shimizu and Hitt, 2004). Thus, the vision of top management on supply chain agility is important in enabling supply chain agility. This study argues that top management needs to participate actively in strategy and planning, with a focus on turning market challenges into competitive opportunities in a timely and cost-efficient manner. Moreover, top management needs to work proactively across business units and organizations to identify opportunities and use inter-organizational collaboration as one of the levers for marketing strategy. The above-mentioned competence implies that top management focuses heavily on inter-organizational collaboration in strategy formulation and planning. This helps the organization have a better understanding of market situations, and thereby be able to identify market changes and respond in a timelier manner. Based on the above argument, we present the following proposition:

Proposition 4a. *Role and vision of top management are positively associated with supply chain agility.*

3.3.2. Competence of employees

The role and vision of top management secure the importance of supply chain agility for driving organizational competitive advantages, whereas the competence of employees represents the ability to implement the organizational response to market changes. With reference to the competence required for organizational agility (Fink and Neumann, 2007), this study argues that the necessary competencies for the implementation of supply chain agility related to strategy or plan include (1) adequate knowledge of the different functions of the supply chain and understanding of the complete business environment; (2) the ability to communicate effectively with different supply chain parties, working in collaborative environments, and implementing inter-organizational projects; and (3) the ability to integrate the supply chains of different organizations or business units and identify ways to enhance the efficiency and effectiveness of responses to market situations. Through the above-mentioned management competencies, the organization is equipped with valuable human resources able to sense market changes, master inter-organizational collaboration, and enhance supply chain agility through the ability to integrate supply chains and improve supply chain operations. Therefore, we offer the following proposition:

Proposition 4b. *Competence of employees is positively associated with supply chain agility.*

Fig. 2 shows the research framework of this paper. This study argues that IT, operational, and management competencies are critical enablers of supply chain agility, which, in turn, is positively related to firm performance.

4. Research methodology

This study is exploratory in nature and no statistical analysis is performed. The sample for this project consists of business units of three fashion and textile companies in Hong Kong, one of the world’s fashion and textile supply chain centers providing services of design, sourcing, manufacturing, and marketing. The case studies fit the objective of investigating the role of IT, operational, and management competencies in enhancing supply chain agility and firm performance. The multiple case studies approach is a preferred methodology for answering the “how” and “why” questions (Slaughter et al., 2006; Yin, 2003); thus, it is an ideal methodology for this exploratory study. Moreover, multiple case studies enhance the external validity of this research design.

Supply chain management is at the dawn of a new era, and an effective supply chain is considered particularly important in the textile and garment industry. This study selects the fashion and textile industry as the context of analysis because this industry is characterized by short product lifecycles, volatile demand patterns, low predictability, highly impulsive purchases, and complex supply chain (Bruce et al., 2004; Christopher et al., 2004; Quinn et al., 2007). This study follows the purposeful sampling strategy to secure the generalizability of the results (Lyytinen and Rose, 2003) and is made to reflect the

strategy of “literal replication” of Yin (1989). Multiple case studies and the logic of replication are employed to strengthen literal replication and validate the propositions. Three manufacturers of different sizes in the fashion and textile industry are chosen to minimize the potential bias arising from firm size. A semi-structured interview protocol (see Appendix) is prepared and interviews are replicated across organizations to ensure rigorous data collection.

To ensure the inclusion of organizations with different sizes in our sample, convenience sampling is adopted in selecting the three target organizations. The selected organizations are small-to-large in scale, with the number of employees ranging from 430 to more than 47,000. Top management representatives of three different organizations are interviewed based on semi-structured interview protocol. They have extensive experience with organizational strategy, supply chain operation, and business environment of the fashion and textile industry. Two waves of data collection are included. In the first wave, we interview the director of the selected organizations (directors from Companies 1 and 2, and a managing director from Company 3) regarding their experience and perception on how different types of competencies help them achieve supply chain agility and enhance firm performance. In the second wave, we interview the directors regarding their inter-organizational collaboration practices and ask them to further confirm the association between supply chain competence and supply chain agility. The average lengths of the first- and second-wave interviews are 2.5 and 1.2 h, respectively.

4.1. Case analysis

In this section, a brief background of the three companies is provided. Company 1 is a giant manufacturer with more than 47,000 employees. It has a vertical setup for both fabric and garment in more than three countries, including China, Malaysia, and Vietnam. Although the headquarters of Company 1 is located in Hong Kong, it has a global presence, with operations in the US, Europe, Japan, and China. Company 1 has a sound financial background that helps it invest in technologies such as SAP. It has implemented value-adding supply chain programs such as vendor-managed inventory and capacity requirements planning with its key customers. Company 2 is a large garment manufacturer with more than 4000 employees. It has different production sites (i.e., China, the Philippines, and Vietnam) to balance the global sourcing dynamic. It does not own any fabric production assets and needs to source from outside suppliers. Company 2 operates under a typical Chinese family business model. Company 3 is a small, fashion-driven manufacturer with approximately 400 employees. Its operation is housed in China, and its market niche is luxury knit fabric.

4.2. Data analysis

The data analysis for each company consists of two major phases: (1) preparation and transcription and (2) verification of the interview report. To assure its quality and integrity, the phase of preparation and transcription of the interview report consists of the following steps: (i) conduct interviews and create field keyword notes, (ii) transcribe the interviews, and (iii) analyze interview transcripts and identify keywords. At this phase, each author of this study independently codes the competence and identifies the relationships. In the second phase, the integrity and correctness of the transcript contents are verified, the codes of each author's transcript are cross-checked by the other authors. Inconsistencies, unclear or disagreement toward transcription are discussed to reduce the biases of the individual researchers.

5. Discussion of results

In this section, the information from the cases is used to validate the relationship among supply chain competence, supply chain agility, and firm performance in the model. Tables 2–9 summarize the case evidence and interview quotes to substantiate our propositions and the degree of importance of each competence toward each company.

5.1. Effect of operational competence on supply chain agility

Based on the case studies, organizations with different sizes agree that operational competencies (i.e., supply chain integration, supply chain flexibility, and supply chain learning orientation) have a positive effect on supply chain agility. However, they place different values and levels of importance toward each operational competence. Giant manufacturers (Company 1) tend to place a higher value on supply chain integration, whereas smaller scale manufacturers (Companies 2 and 3) tend to value supply chain flexibility to a greater extent.

5.1.1. Supply chain integration

Company 1 reports that different processes of the supply chain support one another. They have clear roles and responsibilities in each function or department. They regularly carry out cross-sectional meetings to foster supply chain integration. For instance, the operation team would give suggestions to the sales team on how to approach customers and create a win/win business relationship. The informant reports that supply chain information sharing is critical to supply chain agility because supply chain visibility is the key to making the right decision in response to customer demands. Moreover, supply chain process integration, with a high extent of involvement with supply chain partners, helps Company 1 outperform competitors by providing effective and efficient flows of services to customers.

Table 2

Effect of supply chain integration on supply chain agility and the degree of importance of supply chain integration to each company.

Proposition	Company	Case evidence	Degree of importance	Degree of support
2a. Supply chain integration is positively associated with supply chain agility	Company 1	– clear roles and responsibilities in each function or department – cross-sectional meeting (operation team would give suggestions to the sales team on how to approach customers and create a win/win business relationship) – supply chain information sharing is critical to supply chain agility, as supply chain visibility is the key to making the right decision in response to customers' demands – supply chain process integration, with a high extent of involvement with supply chain partners, helps Company 1 outperform competitors by providing effective and efficient flows of services to customers	Strong	Partly supported
	Company 2	– “information of all the teams, including sales, purchasing, logistics, finance, and so on, is put on the ERP systems as well as the Intranet” – maintains a very close relationship with the supply chain partners, in that it counts on comments from its customers to enhance the supply chain process – inter-organization collaboration in supply chain integration helped Company 2 to understand the requirement of customers and to respond faster to those demands	Medium	
	Company 3	– “information certainly is important to me; I need the information to understand the actual situation and make right decisions” – scale is too small that supply chain integration seems to have insufficient contribution to maintaining a high degree of supply chain agility	Low	

Table 3

Effect of supply chain flexibility on supply chain agility and the degree of importance of supply chain flexibility to each company.

Proposition	Company	Case evidence	Degree of importance	Degree of support
2b. Supply chain flexibility is positively associated with supply chain agility	Company 1	– extent of flexibility is not very high – “our company does have tremendous resources, but we are just too big that it takes several days to make any changes in the supply chain” – “I do think this (supply chain flexibility) is a good strategy to help our supply chain agility; unfortunately we are too big to move quickly as some situations require” – “we have plenty of resources, so how to coordinate the resources and create supply chain visibility is critical”	Low	Partly supported
	Company 2	– “my company's cornerstone is built on flexibility” – cannot invest a lot in technology to outperform others nor target at very niche market – emphasize on flexibility to survive	Strong	
	Company 3	– flexibility is the most important thing to survive – “Since my company isn't that big and I can control and monitor everything, I can quickly adjust the production according to the customers' request This gives me an invincible position compare with other bigger companies”	Strong	

Information in Company 2 is fully integrated. The informant of Company 2 reports that “information of all the teams, including sales, purchasing, logistics, finance, and so on, is put on the ERP systems as well as the Intranet.” Company 2 does not have a comprehensive mechanism for supply chain integration; however, it maintains a very close relationship with the supply chain partners, relying on comments from its customers to enhance the supply chain process. For instance, it has implemented the replenishment program and quick-response program to satisfy the requirements of customers. The informant of Company 2 reports that information helps the informant make timely decisions and revise the action when incidents come up. Inter-organizational collaboration in supply chain integration helped Company 2 to understand the requirements of customers and respond faster to those demands.

The scale of Company 3 is relatively small; thus, information flows to the owner of the firm. The informant of Company 3 reports that “information certainly is important to me; I need the information to understand the actual situation and make the right decisions.” However, Company 3 believes that information integration is the foundation of supply chain integration. According to the informant, the company's scale is too small; thus, supply chain integration seems to have insufficient contribution to maintain a high degree of supply chain agility.

Table 4

Effect of supply chain learning orientation on supply chain agility and the degree of importance of supply chain learning orientation to each company.

Proposition	Company	Case evidence	Degree of importance	Degree of support
2c. Supply chain learning orientation is positively associated with supply chain agility	Company 1	– “We have a strong culture in learning, training and development Supply chain learning is apparently a long-term strategy for our company to advance and enhance our supply chain agility”	Strong	Partly supported
	Company 2	– do not have resources to perform continuously learning – “Learning will drive up supply chain agility, but I am afraid that my company does not have resources but I am open-minded to any new supply chain technologies or ways of operations”	Low	
	Company 3	– do not have resources to perform continuously learning	Low	

Table 5

Effect of IT integration on supply chain agility and the degree of importance of IT integration to each company.

Proposition	Company	Case evidence	Degree of importance	Degree of support
3a. IT integration is positively associated with supply chain agility	Company 1	– dedicated IT team to support business integration – “IT support is an essential part of our company in process integration because our size is so big. Without the IT infrastructure, there's no way we can link across the company. . . IT integration makes the entire company quickly move towards the same goals and directions. Also, it creates the supply chain visibility across companies” – IT support is so strong that it facilitates much information sharing along the supply chain and customization of information systems to fit business needs	Strong	Partly supported
	Company 2	– “Strong information integration within the company is very critical to us . . . I have 4000 workers and have coverage in China, the Philippines, and Vietnam. Without an IT platform, I cannot disseminate information efficiently. That's why I have intranet, ERP, and a network to facilitate the connection between functions in different countries” – IT integration helps the organization “run faster . . . creates visibility so that I can make good decisions. For example, if the Philippine factory is running faster than that in Vietnam as scheduled, I can re-deploy some of the orders from one country to another to serve the customer better. With IT support, we can eliminate a lot of redundant works . . . we can count on EDI for a lot of data transmission activities internally or externally with our customers. This is a critical step to better our supply chain agility”	Strong	
	Company 3	– “Since we are so small . . . I can integrate all different departments if necessary. I think I am doing the same as a sound centralized IT system. However, I do believe the importance of having a sound IT infrastructure for supporting supply chain agility, but it is just not suitable in our situation due to our size”	Low	

Table 6

Effect of IT flexibility on supply chain agility and the degree of importance of IT flexibility to each company.

Proposition	Company	Case evidence	Degree of importance	Degree of support
3b. IT flexibility is positively associated with supply chain agility	Company 1	– “contributes significantly to our supply chain agility” – “IT flexibility is the core part of our company; we cannot function without a good IT connectivity . . . Our company has an excellent IT team to quickly add or modify IT infrastructure depending on the business needs so that we can respond quickly to address to customers' needs . . . the senior management always looks for a better system to strengthen its relationship with customers or better our staff's efficiency”	Strong	Partly supported
	Company 2	– contribute to supply chain agility – “IT flexibility helps me as well as my management team to gain supply chain visibility, act quickly, and respond to the business needs” – “In most of the case it helps, but sometimes I do think I need to control for the budget”	Medium	
	Company 3	– IT flexibility is not an issue of matter	Low	

Table 7

Effect of role and vision of top management on supply chain agility and the degree of importance of top management role and vision to each company.

Proposition	Company	Case evidence	Degree of importance	Degree of support
4a. Role and vision of top management are positively associated with supply chain agility	Company 1	– top management is amenable to investing a huge sum in IT to support supply chain agility – encourages the organization of a cross-sectional meeting where the staff members from different departments meet to discuss the business challenges they encounter and together solve the problems – without the vision and support of the firms' top management in supply chain agility, the IT systems and supply chain integration will not be as successful as it should be	Strong	Fully supported
	Company 2	– top management believes in the value of supply chain agility directly lead to the active participation of employees in building different competencies for supply chain agility	Strong	
	Company 3	– top management believes in the value of supply chain agility directly lead to the active participation of employees in building different competencies for supply chain agility	Strong	

Table 8

Effect of competence of employees on supply chain agility and the degree of importance of employees' competence to each company.

Proposition	Company	Case evidence	Degree of importance	Degree of support
4b. Competence of employees is positively associated with supply chain agility	Company 1	– believes that an employee's competence is one of the key drivers for its efficiency and responsiveness to market changes – employees of different functions are equipped with the necessary competence needed to fulfil supply chain agility – the organization could not achieve a high extent of supply chain agility if its employees are not competent in supporting the strategies and plans of the top management	Strong	Partly supported
	Company 2	– "These competencies are the key drivers that enhance our efficiency and responsiveness to our customers. It helps us differentiate ourselves from our competitors and create a competitive edge"	Strong	
	Company 3	– top management is actively involved in the implementation of strategies as the size of the organization is not big – top management gives concrete directions to his employees and teaches them how to follow his strategies in managing the organization. Therefore, competence of employees seems to play a less important role in organizations with small-scale operations	Low	

Table 9

Effect of supply chain agility on firm performance and the degree of importance of supply chain agility to each company.

Proposition	Company	Case evidence	Degree of importance	Degree of support
1. Supply chain agility is positively associated with firm performance	Company 1	– important for the company to accommodate the needs of the customers – does not only compete in agility but also in cost	Medium	Fully supported
	Company 2	– most important strategy to survive the competitive environment – could not compete through a low-cost strategy – maintaining a high degree of supply chain agility to satisfy market requirements is the only way they can differentiate themselves from large manufacturers	Strong	
	Company 3	– most important strategy to survive the competitive environment – could not compete through a low-cost strategy – maintaining a high degree of supply chain agility to satisfy market requirements is the only way they can differentiate themselves from large manufacturers	Strong	

5.1.2. Supply chain flexibility

The informant of Company 1 reveals, "our company does have tremendous resources, but we are just too big that it takes several days to make any changes in the supply chain. For example, if a customer needs 10,000 pieces of shirts from our China facility in 30 days, we may need to move out some production from China to Malaysia to free up some space for

the quick order. However, it may take a few days from Malaysia to see and confirm if they can accommodate the order. When they confirm back, it becomes too late, and the customer may have already placed the order to our competitors.” Although Company 1 has some degree of flexibility, the flexibility is not high. Nonetheless, the informant says that flexibility is important to obtain supply chain agility. According to the informant, “I do think this (supply chain flexibility) is a good strategy to help our supply chain agility; unfortunately, we are too big to move as quickly as some situations require.” Moreover, coordination among inter-business units is critical for Company 1 in enhancing supply chain agility. “We have plenty of resources, so how to coordinate the resources and create supply chain visibility is critical. In addition to the previous example, the Malaysia business unit may have the production capacity, but I don’t know how I could satisfy the customer’s sudden demands,” the informant states.

Companies 2 and 3, whose operations are smaller in scale, place a higher value on supply chain flexibility. The informant from Company 2 says, “My company’s cornerstone is built on flexibility. . . I am not in the same situation as the large manufacturer who can invest a lot in technology to outperform others. . . unlike the very small one who targets the niche market. We have to emphasize flexibility to survive. . . Compared with other big companies, I can determine and move around my production to accommodate market needs.” Similar to Company 2, Company 3 believes that flexibility is the most important aspect for survival. The informant of Company 3 reports, “Since my company isn’t that big and I can control and monitor everything, I can quickly adjust the production according to the customers’ request. This is what other companies cannot do because they may lose efficiency or cause disruption from impromptu change. . . If my customers call me and make a request, I can respond immediately. This gives me an invincible position compared with other bigger companies. . . My company’s value depends on flexibility and, in return, it drives up the supply chain agility to my customers.” Thus, both Companies 2 and 3 place a high value on supply chain flexibility and believe that flexibility allows them to respond to market changes in a timely manner.

5.1.3. Supply chain learning orientation

In supply chain learning orientation, only the large manufacturer pays attention to its contribution to supply chain agility. The informant of Company 1 says, “We have a strong culture in learning, training, and development. . . Supply chain learning is apparently a long-term strategy for our company to advance and enhance our supply chain agility.” Both Companies 2 and 3 report that they do not have the resources to achieve continual learning but perceive it as important. The informant of Company 2 says, “Learning will drive up supply chain agility, but I am afraid that my company does not have resources. . . but I am open-minded to any new supply chain technologies or ways of operations.” Although smaller organizations do not have the resources to perform continual improvement of supply chain operations, they also believe that learning can increase agility.

The case studies show that organizations with different scales place different values on different operational competencies. A larger organization tends to put a higher value on supply chain integration and supply chain learning, whereas a smaller organization focuses more on supply chain flexibility. The difference is due to the need for and amount of available resources different organizations possess. In larger organizations, supply chain operation is complex and involves many different supply chain parties and business units. Investing more in supply chain integration and supply chain improvement to maintain a high degree of supply chain visibility is therefore necessary. For an organization with a smaller scale of operation, supply chain operation in the organization is simpler. It can coordinate well with supply chain partners without a comprehensive system for supply chain integration. Due to the simple supply chain structure, it can change the supply chain operation or reverse any action to respond easily to a market situation. As a result, it pays more attention to supply chain flexibility in enabling supply chain agility.

The case studies show that organization with different scales place different values on different operational competencies. A larger organization tends to pay a higher value on supply chain integration and supply chain learning, whereas a smaller organization focuses more on supply chain flexibility. The difference is due to the need for and amount of available resources different organizations possess. In larger organizations, supply chain operation is complex and involves many different supply chain parties and business units. It is important for them to invest more in supply chain integration and supply chain improvement to maintain a high degree of supply chain visibility. For an organization with a smaller scale of operation, supply chain operation in the organization is simpler. It can coordinate well with supply chain partners without a comprehensive system for supply chain integration. Due to the simple supply chain structure, it can change the supply chain operation or reverse any action to respond easily to a market situation. As a result, it pays more attention to supply chain flexibility in enabling supply chain agility.

5.2. Effect of IT competence on supply chain agility

IT plays a vital role in supporting supply chain operations. All companies agree that IT integration and IT flexibility are enablers of supply chain agility. The case studies show that the contributions of IT competence to supply chain agility depend on the scale of the organization or complexity of the supply chain.

5.2.1. IT integration

Company 1 has a dedicated IT team to support business integration. The informant says the team is responsible for liaising with all different teams and directly reporting to the supervisor to ensure the proper flow of information. He says, “IT

support is an essential part of our company in process integration because our size is so big. Without the IT infrastructure, there's no way we can link across the company. . . IT integration makes the entire company move quickly toward the same goals and directions. Also, it creates supply chain visibility across companies. It can make everyone work faster, processes change faster, and deliver products faster. There's no question that IT support in process integration in our company plays a vital role in enhancing supply chain agility." The informant reports that the company's IT support is so strong that it facilitates much information sharing along the supply chain and customization of information systems to fit business needs. "During the cross-divisional meeting, the IT team addresses business problems and is very helpful in enhancing our agility," the informant reveals. Moreover, the informant says that IT integration helps the company integrate with its customers. "Without IT support, we cannot integrate with our customers and nor obtain the business in the end."

Similar to Company 1, Company 2 enables supply chain agility through IT integration. According to the informant, "Strong information integration within the company is very critical to us. . . I have 4000 workers and have coverage in China, the Philippines, and Vietnam. Without an IT platform, I cannot disseminate information efficiently. That's why I have intranet, ERP, and a network to facilitate the connection between functions in different countries." The informant reports that IT integration helps the organization "run faster. . . creates visibility so that I can make good decisions. For example, if the Philippine factory is running faster than that in Vietnam, as scheduled, I can re-deploy some of the orders from one country to another to serve the customer better. With IT support, we can eliminate a lot of redundant work. . . We can count on EDI for a lot of data transmission activities internally or externally with our customers. This is a critical step to better our supply chain agility."

The informant of Company 3 says, "Since we are so small. . . I can integrate all different information from different departments if necessary. I think I am doing the same as a sound centralized IT system. . . However, I do believe in the importance of having a sound IT infrastructure for supporting supply chain agility, but it is just not suitable in our situation due to our size."

The case studies show that IT integration helps an organization enhance supply chain visibility, efficiently make changes, and integrate with customers. The researchers of this study are convinced that IT integration is important in enabling supply chain agility. An important point is that IT integration is driven by the scale of the organization or complexity of the supply chain. The larger the organization or the more complex the supply chain is, the higher the value of IT integration to supply chain agility.

5.2.2. IT flexibility

Company 1 reports that IT flexibility "contributes significantly to our supply chain agility." In Company 1, IT infrastructure has the ability to connect with any IT component across the organization and share any form of information. The informant says, "IT flexibility is the core part of our company; we cannot function without a good IT connectivity. . . Our company has an excellent IT team to quickly add or modify IT infrastructure depending on the business needs so that we can respond quickly to address customers' needs. . . Senior management always looks for a better system to strengthen its relationship with customers or better our staff's efficiency." Company 2 shares a similar view with Company 1 on the contribution of IT flexibility. According to the informant, "IT flexibility helps me as well as my management team to gain supply chain visibility, act quickly, and respond to business needs." Again, owing to company scale, IT flexibility is not a problem for Company 3.

IT flexibility helps an organization prepare for unexpected market changes. Based on the case studies, IT flexibility connects every IT component (e.g., internal systems of different departments within an organization or information systems of different organizations) and make IT infrastructure enhancement much easier. Thus, IT flexibility allows the organization to make changes on IT infrastructure efficiently and support any form of supply chain changes. In other words, IT flexibility enables supply chain agility by satisfying any sudden changes in the supply chain.

5.3. Effect of management competence on supply chain agility

The case studies show that the top management of firms, who defines the strategies, and the employees, who implement the strategies, are positively associated with supply chain agility. However, the size of an organization influences the contribution of the competence of employees to supply chain agility.

5.3.1. Role and vision of top management

All the companies believe that top management plays an important role in enabling supply chain agility. In Company 1, top management is amenable to investing a huge sum in IT to support supply chain agility. Moreover, it encourages the organization of a cross-sectional meeting where the staff members from different departments discuss the business challenges they encounter and solve the problems together. Without the vision and support of the top management of firms in supply chain agility, IT systems and supply chain integration would not be as successful as they should be. Due to the size of Companies 2 and 3, most of the decisions are made by the owner. Both companies report that they drive resources to support supply chain agility because they understand that agility brings better business performance. Moreover, they report that their beliefs in the value of supply chain agility directly lead to the active participation of employees in building different competencies for supply chain agility. Therefore, the researchers of this study are convinced that the role and vision of the top management enhance supply chain agility.

5.3.2. Employee's competence

Company 1 believes that an employee's competence is one of the key drivers for efficiency and responsiveness to market changes. In Company 1, employees of different functions are equipped with the necessary competence to fulfill supply chain agility. For instance, the supply chain operation team has substantial knowledge of the whole supply chain operation, whereas an IT team has the ability to integrate the supply chain of different organizations or business units through the development of information systems or deployment of new technology. Moreover, cross-sectional meetings help employees communicate effectively with different supply chain parties. The informant of Company 1 reports that the organization could not achieve a high extent of supply chain agility if its employees are not competent in supporting the strategies and plans of top management. The informant of Company 2 shares a view similar to that of Company 1: "These competencies are the key drivers that enhance our efficiency and responsiveness to our customers. It helps us differentiate ourselves from our competitors and create a competitive edge." In Company 3, top management is actively involved in the implementation of strategies as the organization is not big. According to the informant, he gives concrete directions to his employees and teaches them to follow his strategies for managing the organization. Therefore, competence of employees seems to play a less important role in organizations with small-scale operations.

Based on the case studies, the role and vision of top management have an essential contribution to supply chain agility, regardless of the size of the organization. However, the contribution of competence of an employee is more obvious in large organizations. In an organization with a smaller scale of operation, top management can actively participate and direct the implementation of organizational strategy. Moreover, top management can pinpoint any deviations or mistakes if employees are not sufficiently competent to manage supply chain agility. In a large organization, deviations from the organizational strategy are difficult to determine. Thus, the competence of employees in implementing the strategy contributes to a higher extent to supply chain agility.

5.4. Effect of supply chain agility on firm performance

According to existing literature, supply chain agility helps an organization reduce manufacturing costs, enhance customer satisfaction, remove non-value added activities, and maintain a competitive position in the competitive environment (Botani, 2009; Braunscheidel and Suresh, 2009; Lin et al., 2006; Narasimhan et al., 2006; Swafford et al., 2006a). This study extends the above-mentioned findings by explaining the importance of supply chain agility in organizations with different sizes.

Company 1 is a giant manufacturer. According to the informant, supply chain agility is important for the company to accommodate the needs of customers. Company 1 does not only compete in agility but also in cost. Thus, supply chain agility is only one of the competitive strategies Company 1 uses to generate revenues. Companies 2 and 3 consider supply chain agility as their most important strategy to survive in the competitive environment. Informants of both Companies 2 and 3 report that they are unable to compete through a low-cost strategy. Maintaining a high degree of supply chain agility to satisfy market requirements is the only way they can differentiate themselves from large manufacturers.

Based on the case studies, small-scale manufacturers need greater involvement in supply chain agility in order to compete against larger manufacturers. Although large-scale manufacturers possess more resources to invest in infrastructure for supply chain agility, small-scale manufacturers can obtain supply chain agility through simple IT systems and supply chain operations, close relationships with supply chain partners, and management competences. Therefore, the capability of supply chain agility is not limited to large-scale organizations and does not require huge investments. It helps organizations of any scale to sustain competitive advantages.

6. Conclusion and implications

Supply chain agility is an essential quality for the global supply chain. In this study, a framework for exploring the relationship among supply chain competence, supply chain agility, and firm performance is developed. This study provides a conceptual understanding of supply chain competence and proposes supply chain agility as a competitive capability supported by supply chain competence to achieve global supply chain competitiveness. This study utilizes RBV and case studies to elucidate the mechanism of how and why IT, operational, and management competencies are critical in enhancing supply chain agility. In a number of ways, this study contributes to the growing body of conceptual and empirical literature on supply chain agility. The main lessons learned from this study are summarized in Table 10.

From the case studies, one of the major lessons learned is that the degree of importance of supply chain competence to the organization depends on the scale of the organization. In other words, our proposed propositions on the association between supply chain competence and supply chain agility may be moderated by the complexity of supply chain setting. The supply chain of small-scale organizations is comparatively simpler (e.g., with fewer supply chain partners or lower extent of supply chain diversities) than large-scale organizations. Thus, supply chain of these organizations may not need sophisticated IT competence to support supply chain integration and flexibilities. In addition, due to the differences in the complexity of supply chain between large- and small-scale organizations, large-scale organization needs to pay more attention to supply chain integration, whereas small-scale organizations (with a comparatively simpler supply chain setting) may put more resources on supply chain flexibility. Moderated by the effect of firm scale, IT competence acts as the foundation of

Table 10

Lesson learned from the three case studies.

Lesson learned
1. IT competence is the foundation of supply chain competence, in which IT integration supports a better supply chain integration and flexibility, whereas IT flexibility supports better supply chain advancement 2. Inter-organizational collaboration is important to achieve a complete IT integration along the whole supply chain, which, in turn, supports supply chain integration and flexibility 3. Supply chain competence is composed of three interrelated dimensions: information technology, operational, and management competencies 4. The degree of importance of certain supply chain competencies to the organization depends on the scale of the organization a. IT integration and flexibility, competencies of employees, and supply chain learning orientation are dimensions that are more important to large organizations than to small organizations b. Large organizations tend to pay more attention to supply chain integration, whereas small organizations tend to focus more attention to supply chain flexibility c. Role and vision of the top management are important in driving supply chain competence for supply chain agility

supply chain competence, in which IT integration supports a better supply chain integration and flexibility, whereas IT flexibility supports better supply chain advancement. Moreover, inter-organizational collaboration is important in achieving a complete IT integration along the whole supply chain, which, in turn, supports supply chain integration and flexibility. Finally, the role and vision of top management are important in driving supply chain competence for supply chain agility, regardless of the scale of the organization.

This study contributes to the literature by identifying a set of supply chain competencies, namely, IT competence, operational competence, and management competence, which support supply chain agility for competition amid turbulent business markets. A framework that shows how different competencies contribute to firm performance through supply chain agility is proposed. Moreover, supply chain agility is conceptualized as a kind of competitive capability enabled by supply chain competence (IT, operational, and management competencies). This conceptualization helps distinguish the functional differences between organizational capabilities and supply chain competence. We propose that supply chain capability is directly associated with competitive advantages and firm performance, whereas supply chain competence plays a supporting role to capabilities. By distinguishing the difference between capability and competence, the way to build up organizational capabilities becomes clearer.

From the managerial point of view, this study reveals the importance of supply chain agility in enhancing the performance of a firm. Through the case studies, we can see the importance of the role of supply chain competence in supporting supply chain capability and firm performance. With a deeper understanding of the enabler and effect of supply chain agility, the importance of nurturing IT, operational, and management competencies in supporting supply chain agility and surviving in the competitive environment becomes evident. In other words, organizations that want to gain advantages over competitors through supply chain agility can draw resources toward the creation of IT, operational, and management competencies. For instance, organizations can obtain supply chain agility through the formation of a communication network along the supply chain so that supply chain partners collaborate together in managing intra- and inter-organizational processes. However, this is the first study to explore the role of inter-organizational collaboration in supply chain agility. This study reveals the importance of inter-organizational collaboration in creating IT and operational competencies for supply chain agility. We have proposed that supply chain agility requires both internal and external efforts to achieve. Organizations require support from supply chain partners to respond quickly to market changes. For instance, upstream supply chain collaboration enables a more efficient sourcing channel and supply replenishment process, whereas downstream supply chain collaboration creates a visible feedback channel for service delivery. From the case studies, we have confirmed that companies, to a certain extent, are unable to achieve satisfactory supply chain agility without inter-organizational collaborations. Therefore, organization should no longer focus solely on internal competence development. A hyper-competitive environment encourages supply chain partners to collaborate and develop supply chain and IT integration competence to achieve satisfactory supply chain agility. Therefore, insight from the proposed framework will help supply chain and IT managers strategically plan for the advancement of supply chain agility.

7. Limitations and future research

Similar to other case studies, this study has certain limitations in its findings and conclusions. Data are collected from one informant in each organization. Although the researchers of this study have selected informants with substantial knowledge of the organization and the business environment, the use of only one informant may contain certain biases. To overcome such biases, the informants have been asked to support their words with real business examples. The results obtained from the three companies may not be generally applied considering the size of the sample. However, the results provide valuable strategic implications and insights for managers involved in the enterprise information system in supply chain implementation. The researchers are convinced that this study presents a picture of how and why IT, operational, and management competencies enhance supply chain agility and business performance. The identified constructs and proposed model can be used by academicians as a basis for undertaking further empirical research that tests supply chain agility with other factors.

This study provides a framework for supply chain competence for supply chain agility; however, the interaction effect of the different supply chain competencies on supply chain agility has not yet been empirically tested. The researchers are convinced that an empirical study that investigates the direct effects and interaction effects of the supply chain competencies on supply chain agility and firm performance can contribute to a great extent to the literature.

The concept of supply chain competence is said to have three dimensions: IT competence, operational competence, and management competence. Further research efforts may focus on the development of a set of valid and reliable instruments to measure supply chain competence. Second-order confirmatory factor analyses can be performed to demonstrate the convergent and discriminant validity of the three dimensions of supply chain competence and their contributions to an overall construct of supply chain competence. Proving that supply chain competence is a second-order latent construct composed of three sub-constructs is necessary.

Acknowledgements

The authors are grateful for the constructive comments of the two referees on an earlier version of this article. This research was supported in part by The Hong Kong Polytechnic University under Grant No. G-YH95.

Appendix

A.1. Sample interview questions (first wave)

1. Please comment on the degree of supply chain agility in your organization.
2. With reference to the situation of your organization, please share your experiences/views (better with concrete examples) on the following questions:
 - Please describe the extent of IT integration in your organization.
 - How does IT integration enable supply chain agility in your organization?
 - Please describe the extent of IT flexibility in your organization.
 - How does IT flexibility enable supply chain agility in your organization?
 - Please describe the extent of supply chain integration in your organization.
 - How does supply chain integration enable supply chain agility in your organization?
 - Please describe the extent of supply chain flexibility in your organization.
 - How does supply chain flexibility enable supply chain agility in your organization?
 - Please describe the supply chain learning orientation in your organization.
 - How does supply chain learning orientation enable supply chain agility in your organization?
 - Please describe the top management's role and vision toward supply chain agility in your organization.
 - How does top management's role and vision enable supply chain agility in your organization?
 - Please describe the extent of employee's supply chain-related competence in your organization.
 - How does employee's supply chain-related competence enable supply chain agility in your organization?
 - How does the supply chain agility associate with firm performance?

A.2. Sample interview questions (second wave)

1. Please describe the inter-organizational collaboration situation in your organization, especially in relation to supply chain integration, supply chain flexibility, IT integration, and IT flexibility.
2. How does inter-organizational collaboration support supply chain agility?
3. Are the employees of your organization equipped with the following competencies?
 - a. Capability that involves knowledge of the various functions within the business and ability to understand the overall business environment, which involves inter-organizational collaboration.
 - b. Capability that encompasses interpersonal and management knowledge and skills, such as effective communication among supply chain parties, working in collaborative environments, and planning, organizing, and leading projects, which involve inter-organizational collaboration.
 - c. (For IT personnel only) Technical capability that allows them to effectively identify, integrate, and utilize technical components in the process of developing and maintaining these services, which involve inter-organizational collaboration.
 - d. How does each of the above-mentioned capabilities help enable supply chain agility?

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